

DETERMINATION OF THICKNESS OF A THIN WIRE BY USING PARALLEL FRINGES

Aim:

To determine the thickness of a thin wire or hair of variable thickness by forming parallel fringes.

Apparatus:

Sodium vapour lamp, traveling microscope, thin wire or hair or small piece of paper, black paper, reading lens, two plain glass plates.

Formula:

The thickness of the thin wire is given by,

$$d = \frac{\lambda l}{2\beta}$$

Where

l = the distance of the object from edge of the wedge

λ = Wavelength of the monochromatic light

β = fringe width

Introduction and theory:

When two light waves superimpose, then the resultant amplitude in the region of superposition is different from the amplitude of individual waves. This modification in the distribution of intensity in the sum of the amplitude due to two waves is called the interference. When the resultant amplitude is the sum of the amplitude due to two waves, the interference is known as constructive interference and when the resultant amplitude is equal to the difference of two amplitudes, the interference is known as destructive interference. When a thin object like a small piece of paper or wire or hair is placed between two glass plates, an air film will be formed. When a monochromatic light is allowed to fall on the glass plates, there will be formation of parallel fringes.

The experimental arrangement and ray diagram is shown in below figures.

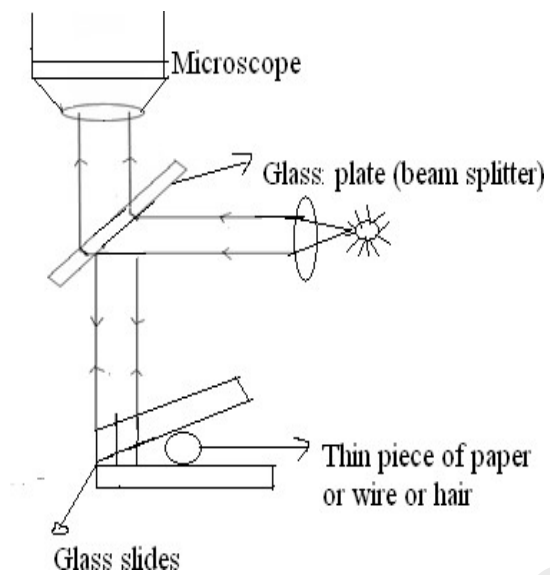


Fig.1: Experimental Arrangement

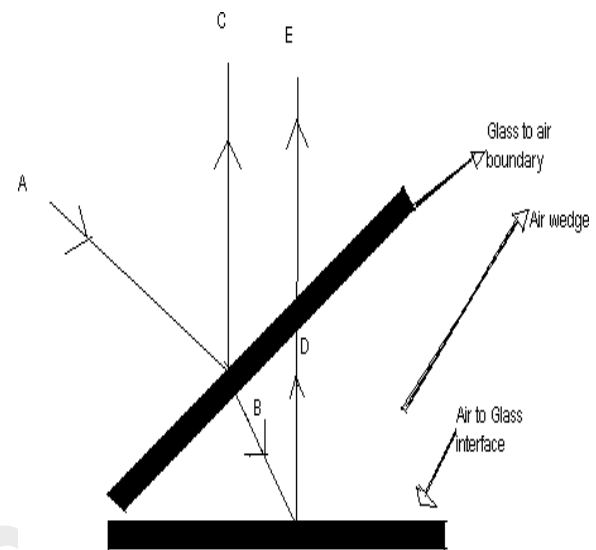


Fig.2: The ray diagram of parallel fringes

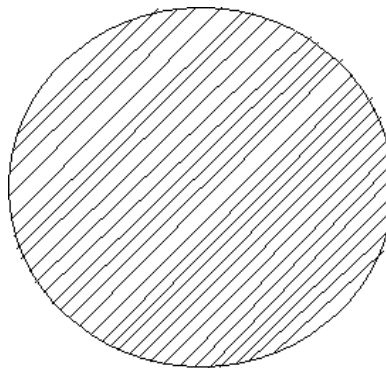


Fig.3: The top view of the parallel fringes

A wedge film of varying thickness having a particular thickness at one end and gradually increasing other thin wedge of air film can be formed. The air film thus formed has a small angle θ . When a light ray AB is incident on such type of air film from above, it partially gets reflected at the glass to air interface (at the top of the air film) in the path BC and part of the transmitted component of the ray get reflected at the air to glass boundary (bottom of the air film) as shown in the ray diagram. The two reflected ray BC and DE are coherent because of the division of amplitude of the original ray AB. The fringe pattern formed is as shown in figure 3. When the light incident is monochromatic, alternating dark and bright fringes are parallel and equidistance from each other will be formed. At the edge of two glass plates the thickness of the air film is zero. There will be formation of destructive interference. So there will be line-Shaped dark fringe at the contact edge. At $t = \frac{\theta}{2}, \frac{3\theta}{2}, \frac{5\theta}{2}$ etc., dark fringes are observed corresponding to the destructive interference.

The fringe width $\beta = 2\tan\theta = 2\theta$ (when θ is small $\tan\theta \approx \theta$)

$$d = \frac{\lambda l}{2\beta}$$

Where l = the distance of the object from edge of the wedge

Procedure:

A small wire or hair is introduced between two glass plates at one end as shown in figure 1. This set up carefully kept on a black paper and placed on the platform of traveling microscope. The air wedge thus formed is illuminated normally by monochromatic light (sodium vapor lamp). The glass plate which is fixed at 45° is helpful in dividing the light into two parts. The microscope is adjusted and focused until the parallel fringes are formed. Starting from one side coincide the cross wires tangential to the parallel fringes and note the reading for every 5 fringes (5^{th} , 10^{th} , 15^{th} etc.).

Table:

Least count of the Travelling Microscope, L.C.=

S.No.	No. of fringes $n = 5$	Reading of the microscope				Width of the n fringes $(x_2 - x_1)$ cm	Fringe width β (cm)
		M.S.R. (a) cm	V.C.	V.C.XL.C (b) cm	Total Reading (a+b) cm		
1	5 th						
2	10 th						
3	15 th						
4	20 th						

Calculations:

Precautions:

1. The microscope should move in one direction from left to right or right to left, so the back Lash error is avoided.
2. To achieve good accuracy in the measurement of 'l' Should be repeated twice or thrice

Result:

The thickness of the thin wire is =